

README FILE FOR THE GLOBAL HISTORICAL CLIMATOLOGY NETWORK hourly (GHCNh)

Version 1.1.0 Updated March 10, 2026

GENERAL INFORMATION

For questions about GHCNh, please email: ncei.orders@noaa.gov.

Information about the GHCNh dataset can be found here:

<https://www.ncei.noaa.gov/products/global-historical-climatology-network-hourly> and
<https://www.ncei.noaa.gov/metadata/geoportal/rest/metadata/item/gov.noaa.ncdc:C01688/html#>

All documentation for GHCNh can be found in:

<https://www.ncei.noaa.gov/oa/global-historical-climatology-network/index.html#hourly/doc/>,
including this README document:
https://www.ncei.noaa.gov/oa/global-historical-climatology-network/hourly/doc/ghcnh_DOCUMENTATION.pdf. Descriptions of additional documentation are shown below.

Version 1.1.0 supersedes all previous GHCNh versions. Differences from the previous version (v1.0.1) are outlined in [ghcnh-version-updates.pdf](#) and column/element changes are shown here: [ghcnh-columns.pdf](#)

The GHCNh data set replaces the legacy “Global Hourly” product, also known as the “Integrated Surface Dataset (ISD)”. For those familiar with ISD, we have provided the matchup between GHCNh and the ISD code groups for ease of reference here in [ghcnh-columns.pdf](#). General information is available here:

<https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database>. The ISD manual can be found here:
<https://www.ncei.noaa.gov/data/global-hourly/doc/isd-format-document.pdf>.

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I. DOWNLOAD QUICK START

Bulk GHCNh station data download access is available in three ways:

1. Period of record station files (data for all years for a single station)
2. Station/year files (one year of data for a single station)
3. A tar file of all station files for a particular year

Data Access Locations	• HTTPS WAF: 1. <a href="https://www.ncei.noaa.gov/oa/global-historical-climatology-network/index.html#hourly/access/by-station/GHCNh_<station>_por.psv">https://www.ncei.noaa.gov/oa/global-historical-climatology-network/index.html#hourly/access/by-station/GHCNh_<station>_por.psv 2. <a href="https://www.ncei.noaa.gov/oa/global-historical-climatology-network/index.html#hourly/access/by-year/<YYYY>/GHCNh_<station>_<YYYY>.psv">https://www.ncei.noaa.gov/oa/global-historical-climatology-network/index.html#hourly/access/by-year/<YYYY>/GHCNh_<station>_<YYYY>.psv 3. <a href="https://www.ncei.noaa.gov/oa/global-historical-climatology-network/index.html#hourly/archive/<archived files>">https://www.ncei.noaa.gov/oa/global-historical-climatology-network/index.html#hourly/archive/<archived files> Where <station> refers to the GHCN identifier • Example file URL: • https://www.ncei.noaa.gov/oa/global-historical-climatology-network/hourly/access/by-year/2023/psv/GHCNh_USW00003812_2023.psv
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For the tar files in the /archive directory which contain all stations per year, the naming convention is as follows:

File Name Pattern	ghcn-hourly_v1.a.b_dYYYY_cYYYYMMDD.tar.gz
File Name Field Definitions	ghcn-hourly = ProductShortName d = data date (i.e., the year provided in the file) c = creation date v1.<a.b> = version number

	YYYYMMDD = date tar = tar file gz = zipped
--	--

The tar file can be untarred and uncompressed by using, e.g., the following Linux command:

```
tar xzvf ghcn-hourly_v1.a.b_dYYYY_cYYYYMMDD.tar.gz
```

ALTERNATIVELY, if you need data for only one or a few specific stations you can:

- Find the station name and identifier in the "ghcnh-station-list" file (e.g., PHOENIX AP [Airport] is "USW00023183"); and
- Then download the data file (i.e., the ".psv" file) for that GHCNh station identifier (e.g., "GHCNh_USW00023183_por.psv" has the data for the whole period of record for the PHOENIX AP) in the /access folder:

<https://www.ncei.noaa.gov/oa/global-historical-climatology-network/index.html#hourly/access/>

Within the

<https://www.ncei.noaa.gov/oa/global-historical-climatology-network/index.html#hourly/access/by-year/><year> directory there are psv and Parquet directories for station-by-year data files.

See Section III on data file formats.

Within the

<https://www.ncei.noaa.gov/oa/global-historical-climatology-network/index.html#hourly/access/by-station/> directory there are entire period-of-record ("por") psv files for each station.

On the main access link,

<https://www.ncei.noaa.gov/oa/global-historical-climatology-network/index.html#hourly/>, along with the /access, /archive and /doc folders, there is an /inventory folder which contains JSON-formatted station inventory files. These are metadata files used by NCEI access tools that facilitate searching by aggregating the minimum and maximum dates for elements within each GHCNh file. These files will likely not be useful for public use.

To view a list of GHCNh station names and geographic coordinates, download the "ghcnh-station-list.txt" or "ghcnh-station-list.csv" from this /doc folder:

<https://www.ncei.noaa.gov/oa/global-historical-climatology-network/index.html#hourly/doc/>

The names and coordinates come from dataset inventories provided by the data source providers. See Section II for details. In addition to the ghcnh station list file, the "ghcnh-inventory.txt" file provides the total number of records by month for the period of record associated with each GHCNh station.

II. Format of the “ghcnh-station-list” Files

The [ghcnh-station-list.txt](#) (and csv) files provide an inventory of the GHCN identifiers and coordinates. The same information is provided in NCEI's [Historical Observing Metadata Repository](#) (HOMR) database.

Variable, Columns (for fixed format), Type

ID, 1-11, Character
LATITUDE, 13-20, Real
LONGITUDE, 22-30, Real
ELEVATION, 32-37, Real
STATE, 39-40, Character
NAME, 42-71, Character
GSN FLAG, 73-75, Character
HCN/CRN FLAG, 77-79, Character
WMO ID, 81-85, Character
ICAO, 87-90, Character

GHCN identifiers are 11 characters beginning with the country's FIPS 10-4 code (2 characters), a network code identifying the station numbering system used (1 character), and finally the station identifier (8 characters). For U.S. stations with a network code of “W” (most airport sites), the last 5 digits of the GHCN station identifier is the WBAN ID. For example, Chicago's O'Hare Airport has a WBAN ID of '94846' and the GHCN identifier is 'USW00094846'.

These variables have the following definitions:

ID=the station identification code. Note that the first two characters denote the FIPS 10-4 country code, the third character is a network code that identifies the station numbering system used, and the remaining eight characters contain the actual station ID.

See [ghcn-countries.txt](#) file for a complete list of country codes.

See [ghcn-states.txt](#) file for a list of state/province/territory codes.

The network code has the following potential values:

A = Retired WMO Identifier used by the USAF 14th Weather Squadron

U = unspecified (station identified by up to eight alphanumeric characters)

C = U.S. Cooperative Network identification number (last six characters of the GHCN ID)

I = International Civil Aviation Organization (ICAO) identifier

M = World Meteorological Organization ID (last five characters of the GHCN ID)

N = Identification number used by a National Meteorological or Hydrological Center partner

L = U.S. National Weather Service Location Identifier (NWSLI)

W = WBAN identification number (last five characters of the GHCN ID)

LATITUDE= latitude of the station (in decimal degrees). North (+); South (-)

LONGITUDE=the longitude of the station (in decimal degrees). East (+); West (-)

ELEVATION=the elevation of the station (in meters, missing = -999.9).

STATE=the U.S. postal code for the state (for U.S. stations only).

NAME=the name of the station.

GSN FLAG=a flag that indicates whether the station is part of the GCOS Surface Network (GSN). The flag is assigned by cross-referencing the number in the WMO ID field with the official list of GSN stations. There are two possible values:

Blank = non-GSN station or WMO Station number not available

GSN = GSN station

HCN=a flag that indicates whether the station is part of the U.S.

CRN FLAG=Historical Climatology Network (HCN) or U.S. Climate Reference Network (CRN). There are three possible values:

Blank = Not a member of the U.S. Historical Climatology
or U.S. Climate Reference Networks

HCN = U.S. Historical Climatology Network station

CRN = U.S. Climate Reference Network or U.S. Regional Climate
Network Station

WMO ID=the World Meteorological Organization (WMO) number for the station. If the station has no WMO number (or one has not yet been matched to this station), then the field is blank.

III. Format of the Data Files

(A) Pipe-Separated Value (.psv) Data Files

Each station "psv" file is made up of a unique identifier and time step followed by 53 variable fields, each with 5 associated metadata/attribute fields (see below for further details and

[ghcnh-columns.pdf](#)). All fields are “pipe” separated (pipe = “|”). Each of the variables and metadata fields are described with explicit names in a header line at the top of each station file.

Pipe-separated (.psv) files can easily be viewed with spreadsheet programs (e.g. Excel) by specifying the delimiter as a pipe “|”.

The variable names and attribute fields are described below:

Each station record is indexed to a single GHCN Station_ID and accompanying Station_name, Latitude, Longitude and Elevation, which were extracted from metadata provided by one or more data sources. Each observation is also indexed to a single date-time in ISO 8601 format, plus the Year, Month, Day, Hour and Minute in Coordinated Universal Time (UTC). Following the identifier, coordinates and time stamp for the observation, there are 53 variables and their accompanying metadata/attributes as follows:

variable (see section IV, Table 1)

variable_Measurement_Code (see section V, Table 2)

variable_Quality_Code (see section VI, Table 3)

variable_Report_Type (see section VII, Table 4)

variable_Source_Code (see section XI, Table 6)

variable_Source_Station_ID (original identifier provided in the data source)

Where *variable* refers to the list of variables in Table 1 and their units. A list of potential Measurement_Codes by variable type is provided in Table 2. The list of Quality_Codes is provided in Table 3. Table 4 provides the list of possible report types. A comprehensive list of column names are provided in Section X “Columns and Headers in PSV Files” below.

(B) Parquet Format Data Files

[Parquet](#) is an open-source file format with easily-indexed columns designed for efficient data storage and retrieval. These files can be read using several languages including Python: <https://arrow.apache.org/docs/python/parquet.html>

IV. List of Elements/Variables

Table 1: Elements/Variables

temperature = 2 meter (circa) Above Ground Level Air (dry bulb) Temperature (°C to tenths)

dew_point_temperature = Dew Point Temperature (°C to tenths)

station_level_pressure = the pressure that is observed at a specific elevation and is the true barometric pressure of a location. It is the pressure exerted by the atmosphere at a point

as a result of gravity acting upon the "column" of air that lies directly above the point.
(hPa)

sea_level_pressure = reduction estimates the pressure that would exist at sea level at a point directly below the station using a temperature profile based on temperatures that actually exist at the station (hPa)

wind_direction = Wind Direction from true north using compass directions (e.g. 360 = true north, 180 = south, 270 = west, etc.). Note: A direction of "000" is given for calm winds.(whole degrees)

wind_speed = Wind Speed (meters per second)

wind_gust = Peak short duration (usually < 20 seconds) wind speed (meters per second) that exceeds the wind_speed average

precipitation = total liquid precipitation (rain or melted snow). Totals are nominally for the hour, but may include intermediate reports within the hour. Please refer to **Section IX "Hourly" Precipitation Measurements** for important details on precipitation totals; a "T" in the measurement code column indicates a trace amount of precipitation (millimeters)

relative_humidity = Depending on the source, relative humidity is either measured directly or calculated from air (dry bulb) temperature and dew point temperature (whole percent)

wet_bulb_temperature = Depending on the source, wet bulb temperature is either measured directly or calculated from air (dry bulb) temperature, dew point temperature, and station pressure (°C to tenths)

pres_wx_MW[1-3] = Present weather observation; MW1 - MW3 is sourced from manual reports; up to 3 observations per report; see [ghcnh-present-weather-elements.pdf](#) (code)

pres_wx_AU[1-3] = Present weather observation; AU1 - AU3 is sourced from automated ASOS/AWOS sensors; up to 3 observations per report; see [ghcnh-present-weather-elements.pdf](#) (code)

pres_wx_AW[1-3] = Present weather observation; AW1 - AW3 is sourced from automated sensors; up to 3 observations per report; see [ghcnh-present-weather-elements.pdf](#) (code)

snow_depth = depth of snowpack on the ground (millimeters)

visibility = horizontal distance at which an object can be seen and identified (kilometers)

altimeter = the pressure "reduced" to mean sea level using the temperature profile of the "standard" atmosphere, which is representative of average conditions over the United States at 40 degrees north latitude (millibars/hPa)

pressure_3hr_change = change in atmospheric pressure measured at the beginning and end of a three hour period; accompanied by tendency code in measurement code field (millibars/hPa)

sky_condition = sky condition observation coverage codes denoting the fraction of the total celestial dome covered by clouds or other obscuring phenomena for the total, total opaque, and total lowest clouds (see [ghcnh-sky-cover-elements.pdf](#) for details) (codes)

sky_condition_baseht = height above ground level (AGL) of the base of the lowest cloud layer and genus types for low, mid and high cloud layers (see [ghcnh-sky-cover-elements.pdf](#) for details) (meters)

ceiling_height = height above ground level (AGL) of the lowest cloud or obscuring phenomena layer aloft with 5/8 or more summation total sky cover, which may be predominantly opaque, or the vertical visibility into a surface-based obstruction; includes determination method and CAVOK (see [ghcnh-sky-cover-elements.pdf](#) for details) (meters)

sky_cover_layer_[1-4] = coverage code denoting the fraction of the total celestial dome covered by up to 4 discrete cloud layers; note this element describes discrete cloud layers, as opposed to the summation of cloud layers as described in the *sky_cover_summation* elements; defines a layer in oktas (i.e. eighths) or tenths of sky covered by cloud (see [ghcnh-sky-cover-elements.pdf](#) for details) (codes)

sky_cover_layer_baseht_[1-4] = discrete cloud base heights at lowest point of layer; up to 4 vertical layers can be reported; corresponds with coverage codes in *sky_cover_layer_1-4* elements (see [ghcnh-sky-cover-elements.pdf](#) for details) (meters)

sky_cover_summation_[1-4] = coverage codes corresponding to the sky cover summation base heights (up to 4 vertical layers can be reported). Each reported cloud layer adds to the total sky coverage and is the summation of cloud layers, as opposed to the discrete layers in *sky_cover_layer* elements (see [ghcnh-sky-cover-elements.pdf](#) for details) (codes)

sky_cover_summation_baseht_[1-4] = height above ground level (AGL) of the base of the cloud layer or obscuring phenomena (meters); up to 4 vertical layers can be reported; corresponds with coverage codes in *sky_cover_summation_1-4* elements; each reported cloud layer adds to the total sky coverage (see [ghcnh-sky-cover-elements.pdf](#) for details) (meters)

precipitation_3_hour = 3-hour total liquid precipitation (rain or melted snow) accumulation from FM12/SYNOP reports; a “T” in the measurement code column indicates a trace amount of precipitation (millimeters); accumulations can be reported over 3, 6, 9, 12, 15, 18, 21 and 24 hours.

precipitation_6_hour = see above

precipitation_9_hour = see above

precipitation_12_hour = see above

precipitation_15_hour = see above

precipitation_18_hour = see above

precipitation_21_hour = see above

precipitation_24_hour = see above

precipitation_5_minute = 5-minute total liquid precipitation accumulation; a “T” in the measurement code column indicates a trace amount (millimeters)

precipitation_15_minute = 15-minute total liquid precipitation accumulation; a “T” in the measurement code column indicates a trace amount (millimeters)

REM = Hourly Remarks present the raw surface observation data in the original format encoded into ICAO-standardized METAR (FM15) or FM12 (SYNOP), FM16 (SPECI) etc format for global dissemination. Further information on decoding these observations can be found in the Federal Meteorological Handbook (FMH) No. 1, Surface Weather Observations & Reports. (Formerly designated as ‘remarks’ in previous GHCNh versions).

V. List of Measurement Codes

Table 2: Element Measurement Codes (where applicable)

Wind Speed / Wind Direction:

A-Abr-Beauf

B-Beaufort

C-Calm

H-5min-avg-spd

N-Normal

R-60min-avg-spd

Q-Squall

T-180min-avg-spd

V-Variable

Atmospheric Pressure Tendency (3-hour pressure change):

Note: in general a 0 through 3 here indicates an increase in pressure over the previous 3 hours and a 5 through 8 indicates a decrease over the previous 3 hours and 4 indicates no change during the previous 3 hours).

0-Incr-then-decr;-atm-pres-same-or-higher-than-3-hrs-ago

1-Incr-then-steady;-or-incr.-then-incr-more-slowly;-atm-pres-now-higher-than-3-hrs-ago

2-Incr-(steadily-or-unsteadily);-atm-pres-now-higher-than-3-hrs-ago

3-Decr-or-steady,-then-incr;-or-incr,-then-incr-more-rapidly;-atm-pres-now-higher-than-3-hrs-ago

4-Steady;-atm-pres-the-same-as-3-hrs-ago

5-Decr,-then-incr;-atm-pres-the-same-or-lower-than-3-hrs-ago

6-Decr,-then-steady;-or-decr,-then-decr-more-slowly;-atm-pres-now-lower-than-3-hrs-ago

7-Decr-(steadily-or-unsteadily);-atm-pres-now-lower-than-3-hrs-ago

8-Steady-or-incr,-then-decr;-or-decr,-then-decr-more-rapidly;-atm-pres-now-lower-than-3-hrs-ag

o

Visibility (code varies by source)

Sources 220, 221, 222, 223, 347, 348

A-Aircraft-horiz-vis

L-Aircraft-slant-rng-vis

M-Max

N-Min

P-Prevailing

S-Sector

Sources: 313, 314, 315, 322, 335, 343, 344,, 346

N-Not-variable

V-Variable

Precipitation (code varies by source)

Sources: 313, 314, 315, 335, 343, 344, 346, 322

1-Measurement-impossible-or-inaccurate

2-Trace

Note: Source 343 precipitation_5_minute only has only 1 possible measurement code: "T" for trace; Source 345 does not report trace precipitation or any other measurement code.

Sources: 220, 221, 222, 223, 347, 348

1-Measurement-impossible-or-inaccurate

2-Trace

3-Begin-accumulated-period-(precipitation-amount-missing-until-end-of-accumulated-period)

4-End-accumulated-period

5-Begin-deleted-period-(precipitation-amount-missing-due-to-data-problem)

6-End-deleted-period

7-Begin-missing-period

8-End-missing-period

E-Estimated-data-value-(eg-from-nearby-station)

I-Incomplete-precipitation-amount,-excludes-one-or-more-missing-reports,-such-as-one-or-more-15-minute-reports-not-included-in-the-1-hour-precipitation-total

J-Incomplete-precipitation-amount,-excludes-one-or-more-erroneous-reports,-such-as-one-or-more-1-hour-precipitation-amounts-excluded-from-the-24-hour-total

Source: 382

Blank = no measurement information applicable

XX = where XX is the number of hours for the precipitation accumulation period.

g = a carry-over measurement flag from the DSI-3240 dataset which was used only on the very first hour of the month if there was zero precipitation during that hour. The purpose of this flag was mainly to indicate that the station was functional and reporting during the month. Normally in DSI-3240, zero precipitation amounts were not included in the data file in order to save space. This HPD dataset does include zero precipitation totals, both those assumed from the DSI-3240 dataset and those determined from the digital recordings of bucket level data.

Z = represents an "assumed" zero precipitation total. Usually these are values from the DSI-3240 dataset. The rule in that dataset was to "assume" a zero total for any hour where nothing else was reported or indicated for that hour as long as the very first hour of the month had a non-zero amount or a zero amount with the "g" measurement flag. Zero amounts were omitted from the DSI-3240 dataset in order to save disk space. We are not concerned with that anymore.

E = Evaporation may have occurred. Data may or may not be reliable.

This flag was used during the period 1984-1993.

T = trace of precipitation

Sky condition and base height, ceiling height, sky cover layers and base heights, sky cover summations and base height layers:

See: (see [ghcnh-sky-cover-elements.pdf](#) for details)

Snow depth (code varies by source):

Sources: 220, 221, 222, 223, 347, 348

1-Unmeasurable

2-Snow-cover-not-continuous

Sources: 313, 314, 315, 322, 335, 343, 344, 346

1-Measurement-impossible-or-inaccurate

2-Snow-cover-not-continuous

3-Trace

4-End-accumulated-period-(data-include-more-than-one-day)

5-End-deleted-period-(data-eliminated-due-to-quality-problems)

6-End-missing-period

E-Estimated-data-value-(eg,-from-nearby-station)

Relative humidity

Sources: 220, 221, 222, 223, 313, 314, 315, 322, 335, 343, 344, 346, 347, 348

D-Derived (values were calculated using air (dry bulb) temperature and dew point temperature)

Wet-bulb temperature

Sources: 220, 221, 222, 223, 313, 314, 315, 322, 335, 343, 344, 346, 347, 348

D-Derived (values were calculated using air (dry bulb) temperature, dew point temperature, and station pressure)

VI. List of Quality Check Codes

Table 3: QC (code varies by source)

A general set of quality control checks is applied to a subset of variables (*temperature*, *dew_point_temperature*, *station_level_pressure*, *sea_level_pressure*, *wind_direction*, and *wind_speed*) after all sources are integrated. These checks are described in the Copernicus [land data user guide](#).

NOTE: For other variables, GHCNh preserves the quality control information of its component (legacy) sources.

The list of codes for the 6 variables mentioned above are:

a : Repeated day streaks

b : Distribution - all

c : Climatological outlier
d : Distribution - monthly
e : Clean up
f : Frequent value
h : High flag rate
i : Precision
k : Repeating streaks
l : Failed logistical consistency
m : Humidity
n : Neighbour comparison outlier
o : Odd cluster
p : Pressure
r : World records exceedance
s : Spike check
t : Timestamp issue
u : Diurnal inconsistency check
v : Variance
w : Winds
x : Excess streak proportion
z : Wind logical - calm masked zero direction

Legacy Codes for Sources 313, 314, 315, 322, 335, 343, 344, 346 include:

0 = Passed gross limits check
1 = Passed all quality control checks
2 = Suspect
3 = Erroneous
4 = Passed gross limits check, data originate from an NCEI data source
5 = Passed all quality control checks, data originate from an NCEI data source
6 = Suspect, data originate from an NCEI data source
7 = Erroneous, data originate from an NCEI data source
9 = Passed gross limits check if element is present
A – Data value flagged as suspect, but accepted as good value.
U – Data value replaced with edited value.
P – Data value not originally flagged as suspect, but replaced by validator.
I – Data value not originally in data, but inserted by validator.
M – Manual change made to value based on information provided by NWS or FAA.
R – Data value replaced with value computed by NCEI software.

Note: for source 343 precipitation_5_minute, there are no legacy source QC codes given.

Legacy Codes for Sources: 220, 221, 222, 223, 347, 348 include

0 Not Checked
1 Good
2 Suspect

- 3 Erroneous
- 4 Calculated
- 5 Removed

Sources: 220, 221, 222, 223, 313, 314, 315, 322, 335, 343, 344, 346, 347, 348

For these sources, relative humidity and wet-bulb temperature are not directly measured, but rather derived from other measured variables. These can have the following QC flags:

- o - Out of range (relative humidity only - with values < 1 or > 100)
- f - Suspect or Error flags for 1 or more of the input measurements

Legacy Codes for Source 382 for precipitation (as given in [C-HPD documentation](#))

Blank = did not fail any quality assurance check

X = failed global extreme exceedance check

N = failed negative precipitation check

Y = failed state extreme exceedance check (performed on daily totals)

K = failed streak/frequent-value check

G = failed gap check

O = failed climatological outlier check

Z = flagged as a result of an official Datzilla investigation

A = The value is not an hourly precipitation total but rather an accumulation total for a period greater than an hour in duration and lasting through the end of this hour. (See measurement flags "a" and "." for the beginning time of the accumulation period and times during the accumulation period, respectively.) Accumulations across multiple hours exist only from the legacy DSI-3240 data source.

M = represents the associated value at this observation time is missing in the DSI-3240 dataset and no alternate data source is available. This is a carry-over indicator from DSI-3240 to allow the user to distinguish between missing and deleted data in that older system. (See the "D" quality flag.) However, the most consistent way to identify hours of missing data across the entire dataset is to test if the precipitation value is equal to the special missing value of -9999.

D = represents the associated value at this time was deleted by the DSI-3240 processing system. Usually this was done manually by a trained meteorological technician who made the decision using ancillary information and experience.

Q = a carry-over quality flag the legacy DSI-3240 data source. Prior to 1996: Indicates value failed an extreme value test (value will be present); data are to be used with caution. Extremes tests were:
1) If the value was not an accumulated precipitation total,

the value failed the one-hour statewide 100-year return period precipitation.

2) If the value was an accumulated precipitation total, the value failed the 24-hour statewide extreme precipitation total. This flag was assigned during a 1997 NCDC rehabilitation of the 1900-1995 DSI-3240 archive.

Since January 1996: A single erroneous datum (value will be present). Lowest data resolution is hourly. This flag is rarely used in DSI-3240 since 1996.

q = a carry-over quality flag the legacy DSI-3240 data source.

Used since January 1996. An hourly value excludes one or more 15 minute periods. Lowest data resolution is 15 minutes.

R = a carry-over quality flag the legacy DSI-3240 data source.

Used since January 1996. Indicates data values are suspect with regard to the times or period of occurrence.

Legacy Codes for Source 345 (for relative humidity and wind speed only):

0 = Good

1 = Issue (field-length overflow)

3 = Error (erroneous data)

VII. Report Type Codes

Table 4: Report type codes

AERO:AERO-Aerological

AUST:AUST-Australia

AUTO:AUTO

BOGUS:BOGUS

BRAZ:BRAZ-Brazil

COOPD:COOPD-USCOOP-SOD

COOPS:COOPS-USCOOP-soiltemp

CRB:CRB-Clim-Ref-Book-from-CDMP

FM12:FM12-SYNOP-fixed-land-stn

FM13:FM13-SHIP-sea-stn

FM14:FM14-SYNOP-MOBIL-mobile-land-stn

FM15:FM15-METAR-Aviation-routine-wx

FM16:FM16-SPECI-Aviation-selected-special-wx

FM18:FM18-BUOY

GREEN:GREEN-Greenland

MESOH:MESOH-Hydro-MESONET-civ-govt

MESOS:MESOS-MESONET-civ-govt

MESOW:MESOW-Snow-MESONET-civ-govt

MEXIC:MEXIC-Mexico
NSRDB:NSRDB-Natl-Sol-Rad-Data-Base
PCP15:PCP15-US-15-min-precip-network
PCP60:PCP60-US-60-min-precip-network
SSA:SAA-Synoptic-airways-auto-merged
SAAU:SAAU-Airways-auto-merged
SAO:SAO-Airways-incl-record-specials
SAOSP:SAOSP-Airways-special-excl-record-specials
SHEF:SHEF-Std-Hydro-Exch-Frmt
SMARS:SMARS-Supp-airways-stn
SOD:SOD-ASOS-AWOS
SOM:SOM-ASOS-AWOS
SURF:SURF-Surf-Rad-Net
SYAE:SYAE-Synop-and-aero-merged
SYAU:SYAU-Synop-and-auto-merged
SYMT:SYMT-Synop-and-METAR-merged
SYSA:SYSA-Synop-and-airways-merged
WBO:WBO
WNO:WNO-WashNavObs

Report type codes for Source 382 hourly C-HPD precipitation
4-DSI-3240
H-derived-HPD-C-high-res

Report type codes for Source 382 15-minute HPD precipitation
6-DSI-3260
H-derived-HPD-R

VIII. Present Weather Code Descriptors

Table 5: Present Weather Codes

Present weather codes are found in <ghcnh-present-weather-elements.pdf>

IX. “Hourly” Precipitation Measurements

For the precipitation totals coming from METARs (Meteorological Aerodrome Reports) from AWOS/ASOS reports, there are usually routine FM-15 messages that are transmitted at a standard time (near the end of the hour: at or after minute 45) for a particular station. In many cases, the report is hourly and in general is no longer than hourly. For the standard hourly reports, there can be METAR SPECI (FM-16) reports transmitted at other (non-standard) times, when conditions change that can affect aviation during the hour between the routine messages.

Some airport sites routinely report more frequently than once per hour (e.g, at 20 minute intervals), even when conditions are not changing. When multiple observations are transmitted within the hour, the precipitation totals represent a running total for that hour. To extract the total precipitation for a particular hour, the last report should be used.

An example of routine (FM-15) and interim/special reports (FM-16) is provided below.

```
METAR KBYY 251255Z 01007KT 7SM -RA BKN020 OVC050 22/21 A2976 RMK A02 T02320216;  
METAR KBYY 251315Z 30008G28KT 3/4SM -RA OVC009 21/20 A2984 RMK A02 P0065 T02230214;  
METAR KBYY 251335Z 35013G18KT 1/2SM +RA BKN005 OVC008 19/19 A2986 RMK A02 P0170  
T01980197;  
METAR KBYY 251355Z 01008KT 4SM RA SCT007 BKN048 OVC060 18/18 A2984 RMK A02 P0186  
T01920192
```

In this case there are two METAR SPECIs after the for hour 13 after the 12:55 UTC routine message. Rainfall amounts are noted by the P groups highlighted above. Precipitation totals are:

P0065 in the 13:15 report means 0.65 inches of rain (16.5 mm) since the last routine METAR (likely since 12:55 UTC)

P0170 in the 13:35 report means 1.70 inches of rain (43.2 mm) since the last routine METAR

P0186 in the 13:55 report means 1.86 inches of rain (47.2 mm) since the last routine METAR

For the intermediate reports:

P0170 at 13:35 UTC indicates that an additional 1.05 inches/26.7 mm (1.70" - 0.65") of rain fell between 13:15 UTC and 13:35 UTC.

P0186 at 13:55 UTC indicates that an additional 0.16 inches/4.1 mm (1.86" - 1.70") of rain fell between 13:35 UTC and 13:55 UTC.

The most straightforward way to extract the hourly total for stations with METAR reports like this one is to take the last report for the hour. The standard reporting time varies somewhat from station to station, but is obvious as the most common minute closest to the end of the hour.

There are multiple sources that contain METAR observations. For the USA these include 343, 335, 220, 221, 223. Prior to 1996, SA (Surface Airways) and SAO reports were used prior to METAR adoption in North America and globally before around 1981. These reports have similar rules to METAR. The precipitation_Report_Type indicates what report the observation came from.

Information on ASOS/AWOS can be found here:

<https://www.ncei.noaa.gov/products/land-based-station/automated-surface-weather-observing-systems>.

For more information on the meteorological report types refer to the Federal Meteorological Handbook (FMH) No. 1, Surface Weather Observations & Reports.

https://www.icams-portal.gov/resources/ofcm/fmh/FMH1/fmh1_2019.pdf

X. Columns and Headers in PSV Files

Files have 329 columns and include individual columns for Year, Month, Day, Hour, and Minute, as well as DATE in ISO format (example: 2024-01-01T02:00:00). The headers are listed in:

[ghcnh-columns.pdf](#)

This also includes the differences between v1.0.1 and the current v1.1.0, as well as a key to the code groups in the legacy dataset, Global Hourly/ISD.

XI. Data Source Codes and Attributes

Table 6: Data Source Codes

Data source codes and attributes are described in: [ghcnh-source-list.pdf](#)